An AI-Ready Air-to-Ground Channel Model for HAPS–UAV Access Networks

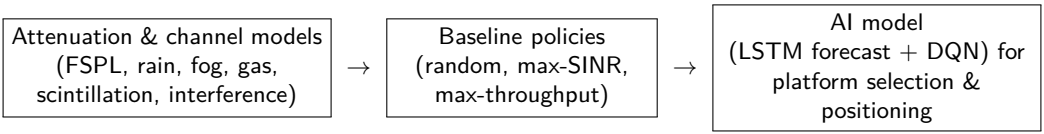
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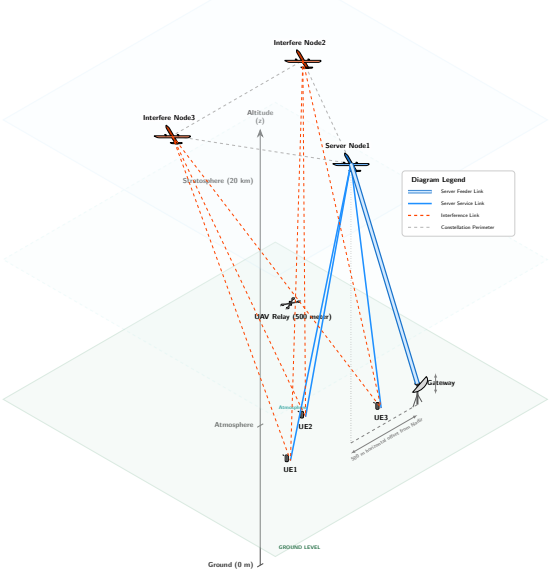
Motivation

- A 3-HAPS equilateral-triangle constellation serves a ground footprint via **service** (2 GHz), **backhaul** (38 GHz), and two-hop **UAV relay** links.
- Distance-only link budgeting misses a key effect: co-channel interference from the other two HAPS makes SINR depend strongly on *angular position*, not just distance.
- Static link-quality analysis cannot react to this, or to time-varying rain fades on the 38 GHz backhaul.
- **AI Ready Model:** a validated channel model, an LSTM rain forecaster, and a DQN agent that learns real-time user–platform association – evaluated honestly, including where it *doesn't* win.

Research flow:



System Topology and Channel Model



Three link types:

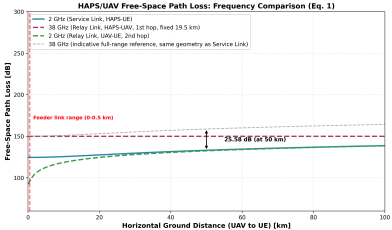
- HAPS $\rightarrow$ UE (service, 2 GHz)
- HAPS $\rightarrow$ Gateway (backhaul, 38 GHz)
- HAPS $\rightarrow$ UAV $\rightarrow$ UE (relay; feeder hop at 38 GHz, UAV$\rightarrow$UE access hop at 2 GHz)

Channel model:

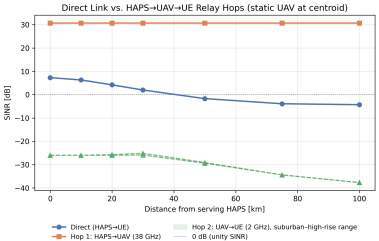
- Free-space path loss (HAPS tier)
- Building-blockage LoS/NLoS probability (UAV tier, Holis & Pechac 2008)
- ITU-R rain, fog, gas, and scintillation losses (backhaul/feeder)

Single relay UAV, fixed at the HAPS triangle's centroid – not hovering beside whichever user it serves (next slide).

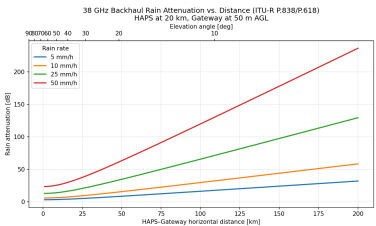
Path Loss and SINR vs. Distance



FSPL vs. distance – same form for 2 GHz and 38 GHz; the feeder sits ≈25.6 dB higher purely from the frequency term.

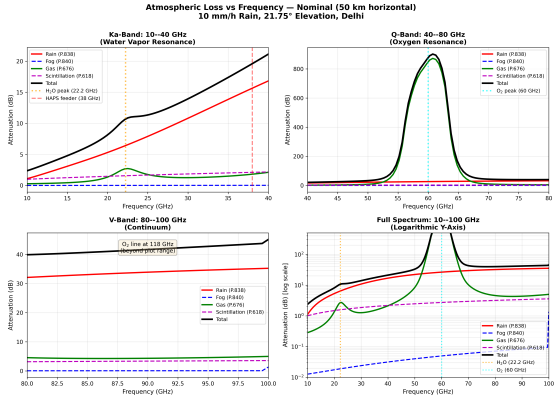


Relay hops vs. distance from serving HAPS – the feeder hop (HAPS→UAV) is flat at 10.2 bits/s/Hz by symmetry; the access hop (UAV→UE) collapses within a few km of the UAV's one fixed position (next slide).



Rain attenuation vs. distance (38 GHz backhaul) – grows faster than linear past ≈100 km as elevation drops and the rain slant-path lengthens.

Why 38 GHz: Frequency Selection



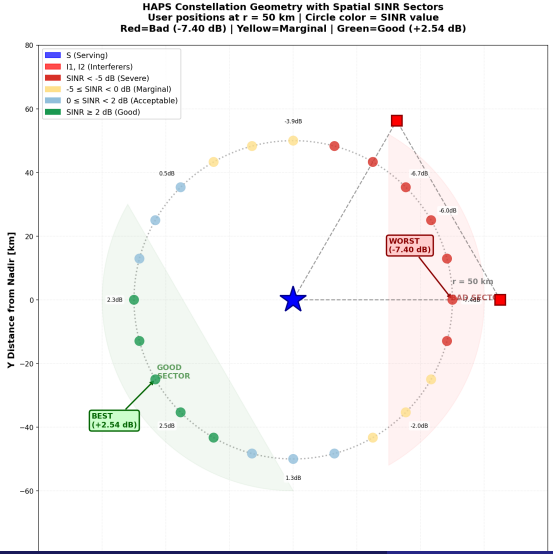
Atmospheric loss (rain, fog, gas) swept across the Ka/Q/V bands for the Delhi gateway path, 10 mm/h rain:

- **Rain** (P.838/P.618) – dominant term, grows roughly linearly with frequency; up to ≈19 dB at 99.9% availability ($R_{0.1} = 15.7$ mm/h).
- **Fog/cloud** (P.840) – small except in dense fog; negligible below 10 GHz.
- **Gas absorption** (P.676) – two resonance peaks to avoid: H_2O at 22.2 GHz, O_2 at 60 GHz (there, total loss exceeds 800 dB over 50 km – unusable).

38 GHz sits in the valley between these peaks (≈0.2–0.3 dB/km clear-sky gas loss).

The rain margin is not design, it's parameters: with literature-sourced hardware constants (43 dBm Tx power, 13.8 dBi antenna gain), clear-sky backhaul SINR is ≈+38 dB – enough to absorb the ≈19 dB rain loss with ≈18.6 dB left over. This is a realistic margin (15–25 dB is typical, ECC Report

Angular Interference Sectors and Scintillation Fading



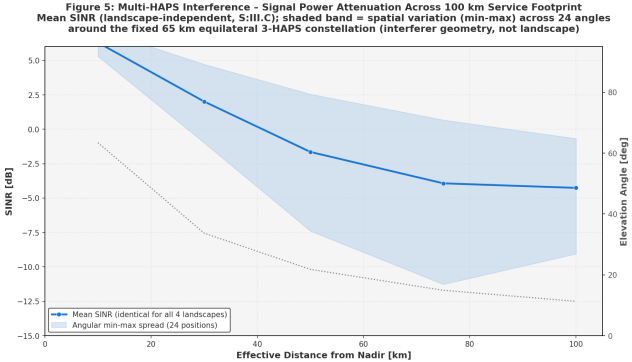
Angle-dependent SINR (sectors): the 3-HAPS triangle creates a good sector (≈120°–290°, away from interferers) and a bad sector (≈330°–75°, toward them) – same radius, up to 9.94 dB apart.

Scintillation (ITU-R P.618): rapid amplitude fluctuations from tropospheric turbulence; grows with frequency, shrinks with elevation. At 30° elevation, $p = 0.01\%$: ≈0.49 dB at 38 GHz vs. ≈0.09 dB at 2.1 GHz.

Favors positioning the UAV relay near nadir (higher elevation) when feasible, especially for the HAPS–UAV feeder hop – which also tends to be away from the interferer-facing sector.

Nuance: each HAPS could simultaneously be a *serving* node (near-nadir, low-loss for its own users) and an *interferer* for the other two HAPS' users – so this isn't a single clean optimization across the constellation (future work, see Conclusion).

Why Learning is Needed: Spatial SINR Heterogeneity



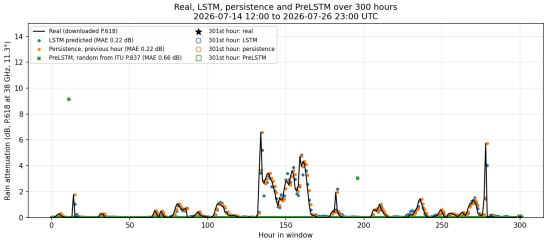
- At a **fixed 50 km radius**, SINR varies by **up to 9.94 dB** depending on angular position relative to the two interfering HAPS.
- For comparison, simply *doubling* the link distance only costs about 2–3 dB – so angular position affects SINR more than distance does.
- A simple "always pick max-SINR" rule cannot anticipate these angular patterns or adapt as weather and battery state change.
- **Conclusion:** a learned, real-time association policy is worth testing – next slides show what it actually buys us.

The Relay UAV: One Static Platform, Not a Follower

A key modeling correction: the relay UAV is fixed at the HAPS triangle's centroid (circumradius $\approx$37.5 km, elevation $\approx$27.5°) – it does *not* hover beside whichever user it currently serves. One UAV cannot be everywhere at once.

- **Feeder hop (HAPS$\rightarrow$UAV):** constant geometry $\Rightarrow$ constant capacity, 10.2 bits/s/Hz, for every HAPS by the constellation's symmetry.
- **Access hop (UAV$\rightarrow$UE):** depends on the user's *real* distance from that one fixed point – capacity falls from 11.1 bits/s/Hz at the UAV's own location to 0.96 bits/s/Hz at 5 km and 0.17 bits/s/Hz at 10 km. Usable radius (capacity $>$ 0.5 bits/s/Hz): $\approx$6.5 km.
- Averaged over azimuth, relay capacity does not exceed the direct link's at *any* distance tested – a single static UAV helps only the minority of users near it, not every user.
- Full, gap-free coverage of a 100 km service radius would need $\sim$200–270 such UAVs (a covering-number estimate) – a fleet-assignment problem, not a scaled-up version of today's single-UAV model (future work).

LSTM Rain-Attenuation Forecaster



- DQN must decide at hour t *before* that hour's rain is observed $\Rightarrow$ needs a forecast.
- Evaluated **out-of-sample** (2026 forecasts from a model trained only on 2023–2025 data).
- Properly **early-stopped**: validation (2025) RMSE is best at epoch 4, not 30 – training longer overfits (next slide).

- LSTM predicts feeder (38 GHz) rain attenuation up to **six hours ahead** from NASA POWER / METAR / ERA5 weather history for Delhi.
- At this validated stopping point, the LSTM beats persistence on RMSE at every horizon from $+2$ h onward (Table, next slide); persistence only wins at $+1$ h.
- A validated, per-horizon-weighted ensemble of LSTM + persistence (weight chosen on 2025, applied once to 2026) adds a small further gain at the shortest horizon.
- **But:** a more accurate forecast $\neq$ a forecast the DQN can use – see the ablation result ahead.

LSTM vs. Persistence, 2026 Test Set

Horizon	LSTM MAE	Persist. MAE	LSTM RMSE	Persist. RMSE
+1 h	0.15	**0.12**	0.54	**0.53**
+2 h	0.21	**0.20**	**0.70**	0.75
+3 h	**0.24**	0.26	**0.80**	0.90
+4 h	**0.27**	0.30	**0.86**	1.02
+5 h	**0.28**	0.34	**0.90**	1.11
+6 h	**0.29**	0.36	**0.92**	1.17

- 6556 forecast hours, 2026, no 2026 data in training.
- Persistence only wins at the shortest horizon (+1 h) – little changes in an hour, so "tomorrow looks like now" is hard to beat.
- The LSTM's advantage grows with horizon as its learned dynamics start to matter more than the last observed value.

DQN Association Agent: Problem Formulation

Per-user MDP – each user independently chooses a platform; network reward is computed after all assignments.

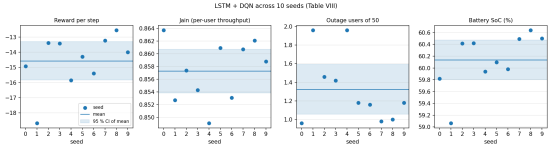
- **State:** SINR to each platform, distance, battery SoC, rainfall, visibility, time-of-day, six LSTM rain forecasts.
- **Action:** $a_k \in \{0, \ldots, 5\}$ – direct link to HAPS1–3, or relay through the one UAV served by HAPS1–3.
- **Reward:** throughput (rain-derated) + fairness (Jain index) − energy cost − outage/degraded-service penalties − low-battery penalty.
- Custom Deep Q-Network implemented directly in PyTorch (`torch/torch.nn`).
- Trained on **2400** rolling 24-hour episodes (400 real 2024 episodes, cycled six times – 400 alone left reward still improving noisily); evaluated on 50 held-out 2025 days, **ten** experiments (random seeds).

Results: Policy Comparison

Policy	Reward/step	Jain	Outage/50
Random (six actions)	−91.9 (1.5)	0.262 (0.005)	41.5 (0.3)
Max-SINR (direct only)	−13.8 (1.7)	**0.865** (0.002)	0.8 (0.2)
Max-throughput (six actions)	**−13.5** (1.6)	0.862 (0.002)	**0.7** (0.2)
LSTM + DQN	−14.6 (1.8)	0.857 (0.005)	1.3 (0.4)

- Mean ± std. dev. over ten experiments, 50 held-out 2025 days.
- With the UAV-relay geometry corrected (previous slides), the DQN **no longer has the highest reward** – it is statistically indistinguishable from, and slightly behind, both rule-based baselines.
- Max-SINR and Max-throughput are both **oracle-style upper bounds**: perfect, instantaneous SINR/capacity knowledge, no learning, no real deployability constraint. The DQN was never expected to beat them – the honest question is what it gets right that they can't (two slides ahead).

Results: Stability Across Experiments



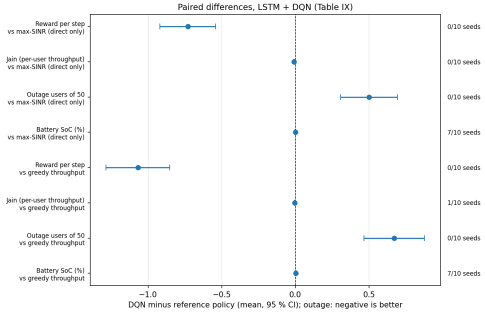
LSTM + DQN, ten independent seeds (dot = one training run; line = mean; band = 95% CI *of the mean*):

- Reward/step: -14.57 ± 1.78 (95% CI $[-15.84, -13.30]$)
- Jain fairness: 0.857 ± 0.005
- Outage: 1.33 ± 0.37 of 50 users
- Battery SoC: 60.14 ± 0.47 %

CI vs. spread: the shaded band is *not* "most runs land here" – it narrows with more seeds even if run-to-run variability stays the same. It shows how precisely the *true mean* is known from these ten runs.

Takeaway: the tight spread across all ten seeds shows the policy is **reliably learned to this level**, not a lucky or unlucky single run – consistency, not the reward level itself, is what this figure demonstrates.

Results: Statistical Significance



Paired differences, LSTM+DQN minus baseline (per experiment, 50 days each, ten seeds). Whiskers crossing 0 = not significant.

vs. Max-SINR:

- Reward: -0.73 ($p < 0.0001$) – DQN worse, 0/10
- Jain: -0.0079 ($p = 0.0003$) – DQN worse, 0/10
- Outage: $+0.50$ ($p = 0.0002$) – DQN worse, 0/10
- Battery SoC: ≈ 0 ($p = 0.19$) – tie

vs. Max-throughput:

- Reward: -1.07 ($p < 0.0001$) – DQN worse, 0/10
- Jain: -0.0043 ($p = 0.0099$) – DQN worse, 1/10
- Outage: $+0.67$ ($p < 0.0001$) – DQN worse, 0/10
- Battery SoC: ≈ 0 ($p = 0.09$) – tie

Takeaway: the DQN loses to both oracle-style baselines on reward, fairness, and outage on nearly every seed, and ties only on battery. The gap traces almost entirely to a higher severe-outage rate (2.5 % of user-hours vs. 1.3–1.7 %) – decomposed next.

What the DQN Gets Right That the Baselines Can't

- Max-SINR needs only a real-time SINR reading – already a deployable rule. Max-throughput needs perfect, instantaneous, non-causal knowledge of every option's capacity including relay – not deployable, and since relay rarely wins after the geometry fix, it collapses in practice to "pick the best direct option" anyway.
- **The DQN is the only policy that retains and actually uses the relay path.** It invokes relay specifically for users whose best direct SINR already averages $-1.9\,\text{dB}$ (population mean $+2.4\,\text{dB}$) – consistent across all ten seeds ($p = 2.4 \times 10^{-13}$).
- Neither rule-based baseline can replicate this: Max-SINR cannot use relay at all; Max-throughput's instantaneous rule has no notion of *which* users most need it.
- So the DQN's value is **structural** (targeted relay fallback for near-outage users), not a reward-table win against oracle-style baselines that were never realistic competitors.

Does the Rain Forecast Help the DQN Itself?

A forecast ablation (`lstm` vs. `none` – forecast features zeroed – vs. `oracle` – true future attenuation), same ten-seed protocol:

- **Result:** `lstm`, `none`, and `oracle` are statistically indistinguishable on reward, fairness, and outage (all $p > 0.1$) – even against the oracle ceiling.
- This is **not** evidence against forecasting in general, or against this forecast's own accuracy (verified two slides back) – it means this action space gives the agent no rain-reactive lever to exploit.
- **Why:** rain already shows up in the state via its effect on current SINR. And both links the agent can act on are already largely insulated from rain – 2 GHz service/access hops are physically resilient to it; the 38 GHz feeder's $\approx$19 dB rain loss is already absorbed by $\approx$18.6 dB of design margin (realistic hardware parameters, not something we tuned).
- A forecast only pays off where an action lets the agent move *ahead of* degradation – this fixed-geometry, six-action MDP doesn't offer that lever yet.

Conclusion and Future Work

Channel model. Spatial SINR heterogeneity (up to 9.94 dB at fixed 50 km) dominates service-link quality; backhaul remains noise-limited. **Relay path.** A single static UAV at the HAPS-triangle centroid has a flat feeder hop but a collapsing access hop – it helps only users within ≈6.5 km, not every user; full coverage would need ∼200–270 such UAVs. **Learned association.** The LSTM-fed DQN is statistically tied with (slightly behind) oracle-style Max-SINR/Max-throughput baselines on reward – but is the only policy that targets relay use at users who actually need it. The rain forecast adds no measurable reward benefit in this action space, without that being evidence against rain forecasting itself.

Future work:

- **Model multiple UAVs:** HAPS-to-UAV assignment as its own fleet-scale MDP, decoupled from per-user access; relay-vs-direct as an explicit policy layer; relative motion of UAVs and HAPS; fleet placement that avoids blind corners.
- **LSTM improvements:** alternative architectures (ConvLSTM, GRU, spatio-temporal GNNs for irregular future sites); the validated ensembling gain already found; forecasting a different impairment if time-series data for it existed (fog, scintillation, gas) – rain was the only one we had multi-year data for.
- **Dynamic topology and geometry:** changing HAPS roles (serving vs. interferer); joint UAV/HAPS positioning across both roles at once; beamforming between HAPS and Gateway.

Thank you – Questions?